

A Counterexample to Bidual Stability of Uniform Property (S)

Automated Math Discovery Smoke Test

June 8, 2026

Source and Open Question

This packet concerns Open Problem (5) from:

William B. Johnson and Tomasz Kania, *Uniform Property (S)*, arXiv:2602.09106.

The source location is the open-problems section of the paper.

(5) If X has uniform property (S), must X^{**} have it too? The converse follows from the principle of local reflexivity.

Problem (5) asks:

If X has uniform property (S), must X^{**} have it too?

We give a negative answer, in the real Banach-space category.

Definitions

For a normed space E , the paper defines

$$U_E(x, y; a) = \sup_{\|z\| \leq a} | \|x + z\| - \|y + z\| |$$

for $x, y \in S_E$ and $a > 0$, and

$$U_E(d; a) = \inf_{\substack{x, y \in S_E \\ \|x - y\| \geq d}} U_E(x, y; a).$$

The space E has uniform property (S) precisely when $U_E(d; a) > 0$ for every $0 < d < 2$ and every $a > 0$.

Counterexample

Theorem 1. *There is a Banach space X with uniform property (S) such that X^{**} fails uniform property (S).*

Proof. Let

$$X = L_1[0, 1]$$

over the real scalars. By Theorem 1.2 of the source paper, $L_1(\mu)$ has uniform property (S) whenever μ is atomless. Thus X has uniform property (S).

We now show that X^{**} does not. Since $X^* = L_\infty[0, 1]$, we have

$$X^{**} = (L_\infty[0, 1])^*.$$

Represent the commutative unital Banach lattice algebra $L_\infty[0, 1]$ as $C(K)$ for its Stone/Gelfand compact space K . The space K has at least two distinct points: for instance, the non-trivial idempotent $\chi_{[0, 1/2]}$ gives a non-trivial clopen set in the Stone representation. Hence

$$X^{**} \cong C(K)^* = M(K),$$

the Banach lattice of regular signed measures on K with the total-variation norm, and there exist two distinct Dirac measures δ_s, δ_t in $M(K)$.

Define two positive norm-one measures

$$u = \frac{3}{4}\delta_s + \frac{1}{4}\delta_t, \quad v = \frac{1}{4}\delta_s + \frac{3}{4}\delta_t.$$

Then

$$\|u - v\| = \left| \frac{3}{4} - \frac{1}{4} \right| + \left| \frac{1}{4} - \frac{3}{4} \right| = 1.$$

We claim that

$$U_{X^{**}}(u, v; 1/8) = 0.$$

Let $z \in M(K)$ satisfy $\|z\| \leq 1/8$. Write

$$z = \alpha\delta_s + \beta\delta_t + z_0,$$

where z_0 has no mass at s or t . Then $|\alpha| \leq 1/8$ and $|\beta| \leq 1/8$. In particular, all four numbers

$$\frac{3}{4} + \alpha, \quad \frac{1}{4} + \beta, \quad \frac{1}{4} + \alpha, \quad \frac{3}{4} + \beta$$

are positive. Since the atoms $\{s\}$, $\{t\}$, and $K \setminus \{s, t\}$ are disjoint for the variation norm,

$$\begin{aligned} \|u + z\| &= \left(\frac{3}{4} + \alpha \right) + \left(\frac{1}{4} + \beta \right) + \|z_0\| \\ &= 1 + \alpha + \beta + \|z_0\|, \end{aligned}$$

and similarly

$$\begin{aligned} \|v + z\| &= \left(\frac{1}{4} + \alpha \right) + \left(\frac{3}{4} + \beta \right) + \|z_0\| \\ &= 1 + \alpha + \beta + \|z_0\|. \end{aligned}$$

Thus $\|u + z\| = \|v + z\|$ for every z with $\|z\| \leq 1/8$. Consequently,

$$U_{X^{**}}(u, v; 1/8) = 0.$$

Since $u, v \in S_{X^{**}}$ and $\|u - v\| = 1$, this implies

$$U_{X^{**}}(1; 1/8) = 0.$$

Therefore X^{**} fails uniform property (S). □

Verification Notes

The proof is exact. The script `code/check_atomic_band.py` samples perturbations in the finite-dimensional model $\ell_1^2 \oplus_1 \mathbb{R}^k$ and checks the identity

$$\|u + z\|_1 = \|v + z\|_1 \quad (\|z\|_1 \leq 1/8)$$

for the same atom coefficients. This is only a sanity check; the measure-theoretic calculation above is the certificate.

Limitations and Novelty Check

This packet answers Problem (5) as stated for real Banach spaces. The example is deliberately classical: $X = L_1[0, 1]$ is one of the paper’s basic positive examples, and the failure in the bidual is caused by two Dirac atoms in the measure representation of $(L_\infty[0, 1])^*$.

I have not performed a literature search beyond the source paper and the local pipeline corpus, so the novelty status is “candidate” pending human and literature review.

Human Review: bidual stability of uniform property (S)

Original automatic proof: `solution_packet.pdf`

Checked date: 10 June 2026

Status: Manually checked.

Verdict: Correct.

Human Comments

The solution is correct. Once the idea is mentioned, the argument is quite easy to spot: choose $X = L_1[0, 1]$, so that the underlying measure is atomless and the positive results of the paper apply to X . Thus X has uniform property (S).

The obstruction appears only after passing to the bidual. Since

$$X^{**} = (L_\infty[0, 1])^*,$$

one can represent $L_\infty[0, 1]$ as $C(K)$ for its Stone space K , and therefore

$$X^{**} \simeq C(K)^* = M(K).$$

Equivalently, by the representation theory of abstract AL-spaces, this space may be viewed as an $L_1(\nu)$ -space for a measure ν with atoms. In the Stone representation, the atoms are visible as Dirac measures.

The key point is that once the bidual contains at least two atoms, uniform property (S) cannot hold. The packet exhibits two Dirac measures δ_s, δ_t and then chooses

$$u = \frac{3}{4}\delta_s + \frac{1}{4}\delta_t, \quad v = \frac{1}{4}\delta_s + \frac{3}{4}\delta_t.$$

For sufficiently small perturbations, the two atomic coefficients stay positive, and the total variation norms of $u + z$ and $v + z$ remain equal. This gives

$$U_{X^{**}}(u, v; 1/8) = 0 \quad \text{while} \quad \|u - v\| = 1,$$

so X^{**} fails uniform property (S).

Short Version

The example is $X = L_1[0, 1]$. The space X has uniform property (S) because the measure is atomless. But X^{**} can be represented as a measure space with atoms, and the presence of two atoms gives a direct two-coordinate obstruction to uniform property (S).

Further Question

The proof uses two atoms. It would be interesting to understand whether the paper's positive results, or some variant of them, extend to the case where the representing measure has exactly one atom. The present packet does not settle that one-atom case; it only shows that two atoms are already enough to destroy uniform property (S).

Author Follow-Up

After reaching out to the authors, they indicated that they had already come up with the same example for a new version that has not yet been published. This should therefore be recorded as a correct rediscovery/check rather than as a novel contribution.

Review Outcome

Archive this packet as a correct negative answer to bidual stability of uniform property (S), with the side note that the one-atom case remains a natural follow-up question. Also record that the same example was already known to the authors for a forthcoming version, so this should not be counted as an independent novel contribution.